

Continuous aerial water delivery by vacuum-lift airship

Version 1 · 9 August 2026

Every model figure is cited by key and checked against the simulation that produces it; source figures are cited in place. Nothing described here has been built.

THE RESULT

A buoyant hull that never lands turns aerial firefighting from a sortie problem into a flow-rate problem. Modelled at three scales, the largest class delivers **13,183 tonnes of water an hour**, continuously, for 4.6 kWh a tonne — and gets cheaper per tonne the larger it is, because buoyancy scales with volume and drag with area. This paper is the arithmetic behind that sentence, and the list of what would have to be true for it to survive contact with hardware.

Status. Every quantitative statement below is either an output of the simulation in this repository, cited by key and verified automatically against `research/figures.json`, or a figure from a catalogued source, cited plainly. Nothing has been built. The model is first-order throughout, and §8 lists the thirteen assumptions known to be wrong or unjustified — eleven open, two corrected on 2026-08-09 and kept in place with what they cost.

1. The problem, and why an airship is a plausible shape for it

Aerial firefighting is a duty cycle problem before it is an aviation problem. A very large airtanker carries something like 70 t, releases it in seconds, and then spends 30–90 minutes flying to a base, loading, and flying back. Its instantaneous delivery rate is enormous and its *sustained* rate is set almost entirely by the turnaround. Helicopters with buckets shorten the turnaround by dipping from a nearby lake, and pay for it in payload — a Chinook carries about 10 t.

The gap in the middle is a vehicle that dips like a helicopter and carries like a tanker. Nothing occupies it, and the reason is that lift-per-unit-mass for rotorcraft is bought continuously with power, so payload and endurance trade against each other directly.

Buoyant flight does not have that trade. A hull that displaces its own weight in air holds altitude at zero power, and the marginal cost of carrying more is the cost of building a bigger hull, once. That is the argument for an airship, and it is not a new one: it has been made for cargo, for surveillance, and for high-altitude platforms for a century, and it has generally lost to the fact that the vehicle is enormous, slow, and vulnerable to weather.

Firefighting is a plausible exception on three counts. The mission is short-range and repetitive, so slowness costs little. The vehicle can be uncrewed, so the risk calculus that has historically killed large airships changes. And — the point of this paper — the water source and the fire are usually within tens of kilometres of each other, so the duty cycle is dominated by loading and unloading rather than by transit, which is precisely where a buoyant vehicle wins.

2. Why vacuum rather than helium

A vacuum displaces the same air as helium and weighs nothing instead of 0.169 kg/m^3 , so it is about 14% more buoyant at sea level. It also cannot leak away, cannot be embargoed, and cannot be priced by a supply chain with two producers of consequence.

The reason nobody flies one is structural. A gas envelope is in tension and can be a fabric; a vacuum envelope is in compression and must resist 101 kPa of atmosphere trying to crush it, which means it must be stiff, which historically means it must be heavy. Lana de Terzi proposed it in 1670 and the arithmetic has failed ever since.

The modern case rests on discrete lattice construction — ultralight periodic cellular structures assembled from discrete parts, which reach very low densities at usable stiffness. Jenett et al. (NASA, 2019) is this project's structural precedent, and the honest summary of it appears in §8.1: **their own numbers do not close our mass budget**. Akhmeteli & Gavrilin (2021) reach a positive result for a vacuum balloon by a different route, and their published payload fraction is 0.1 against our implied 0.5.

So the correct statement is: vacuum lift is not obviously impossible any more, and this model assumes a structure that the two best available sources say is not yet available. Everything downstream inherits that assumption.

3. The vehicle

Three classes, geometrically similar, sized by a single safety requirement (§4).

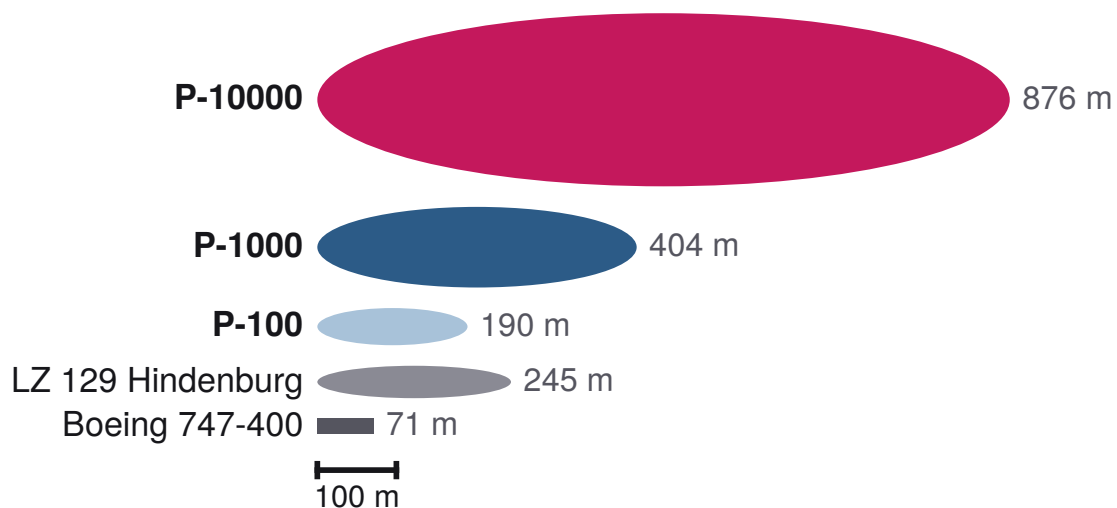


Figure 1. The three hulls at true relative scale, against the largest aircraft and the largest airship ever flown. Drawn from the same lengths and diameters as the table below.

	P-100	P-1000	P-10000
Payload	100 t	1,000 t	10,000 t
Displacement	220,000 m ³	2,200,000 m ³	22,000,000 m ³
Length × diameter	190 × 47 m	404 × 102 m	876 × 219 m
Cruise	90 km/h	110 km/h	130 km/h
Rotors / disc area	4 / 2,500 m ²	6 / 12,000 m ²	14 / 160,000 m ²
Battery	20 MWh	120 MWh	2,000 MWh
Fill rate	0.5 m ³ /s	3 m ³ /s	15 m ³ /s
Descent bag	125 t	1,250 t	12,400 t

Dry mass is set equal to payload in every class. That is an assumption, not a result, and §8.1 is about it.

4. What sizes the hull: fail-safe float-up

The displacement is not chosen to make the numbers work. It is chosen so that a ship that cannot release its water still rises.

*A hull must be positively buoyant at its working altitude while **fully loaded with water** and unable to drop it.*

An uncrewed vehicle whose outlets jam must not sink into the fire it is fighting. Making that a sizing constraint rather than a design goal means the failure mode of a stuck valve is an inconvenient ascent rather than a crash.

The arithmetic is identical per tonne of payload across the classes because dry mass equals payload. Loaded mass is 2 t per tonne of payload; air at the 2,500 m MSL working ceiling is 0.9569 kg/m³; a 5% margin therefore wants 2,194.7 m³ of displaced air per tonne. Published: 2,200 m³/t, giving +5.25% — 21,050.9 t of lift against 20,000 t of loaded ship.

This is where a correction made on 2026-08-09 belongs. The ledger previously bought its lift at sea-level density and spent it at altitude, which overstated lift by 22%. Correcting it grew every hull by 22.2% and, in doing so, created the descent problem in §6 — the buoyancy that guarantees the ship comes up is exactly the buoyancy that stops it going back down.

The two extremes are not the same case, and the model evaluates both:

- **Float-up is hardest at the ceiling**, 2,500 m MSL, in the thinnest air.
- **Descent is hardest at the water**, 1,300 m MSL, in air 16% denser.

Nothing else in this model changes conclusions as often as remembering which of those two applies.

5. The cycle

Six phases, timed rather than asserted. For the P-10000 at 15 km one way:

Phase	Minutes	What happens
SOURCE_APPROACH	5.00	arrive over the lake, come to a dead stop , lower the anchor and pumps
WATER_FILL	11.11	pump at 15 m ³ /s, no yaw (the lines would tangle)
OUTBOUND_TRANSIT	8.14	climb and run to the fire at 130 km/h
WATER_RELEASE	11.11	one long pass, 450 m AGL
BUOYANCY_ESCAPE	2.00	rise off the line on buoyancy alone, no propulsion
RETURN_TRANSIT	8.14	run back, make nitrogen, let down onto the lake
total	**45.51**	10,000 t delivered

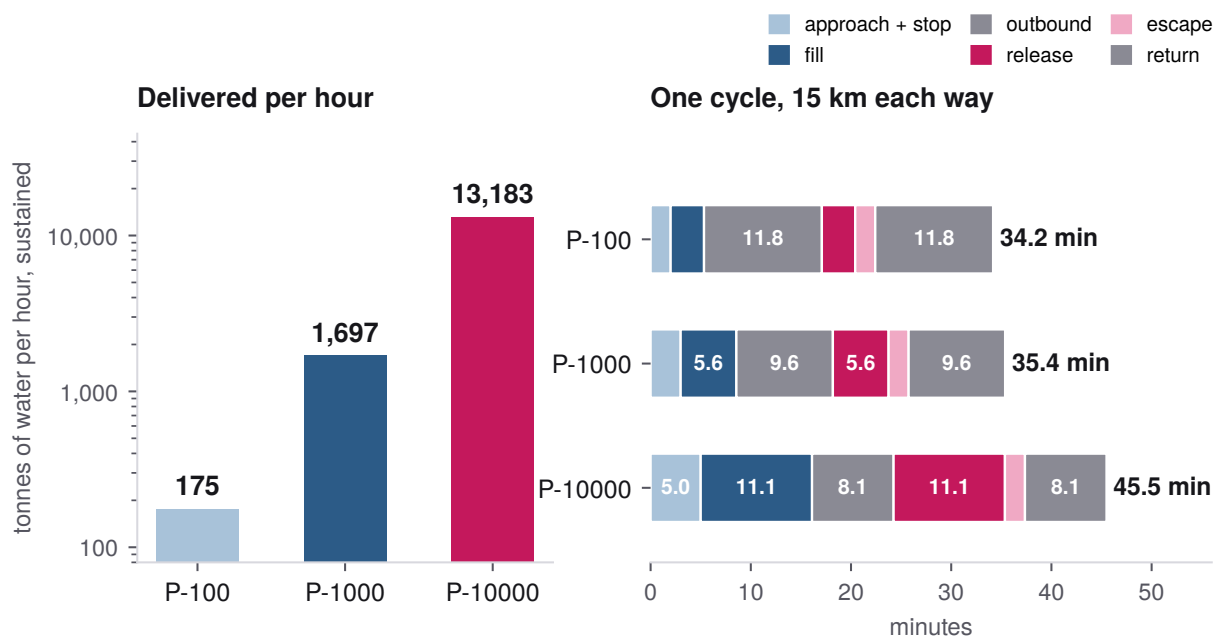


Figure 2. Sustained delivery, and the cycle behind it. Water handling is the longest part of every cycle, which is the signature of a vehicle that never lands.

That is 13,183 t/h and 1.32 drops per hour. Water handling — fill plus release — is 22.2 of the 45.5 minutes, and the model reports the bottleneck as transit distance at this range because both scale but only transit scales with distance.

The drop is one pass, not three. Three circuits meant two turns of an 876 m hull over a fire, which is a manoeuvre the model had no business assuming. A single long release removes them.

6. The descent problem, and the lake as its solution

6.1 The problem

Buoyancy that guarantees the ship rises when loaded must be overcome when it is empty. Worst at the bottom of the letdown, over the water:

	P-100	P-1000	P-10000
Surplus to hold down at the source	135.6 t	1,366.9 t	13,722.5 t
Rotor capability	267.2 t	1,318.1 t	12,666.2 t
Closes on rotors alone?	yes, $\times 1.97$	no	no

The two larger classes cannot reach their own water source under power. This was discovered by moving the force balance from the ceiling — where it had always been struck, and where it closed — to the place where the descent actually ends.

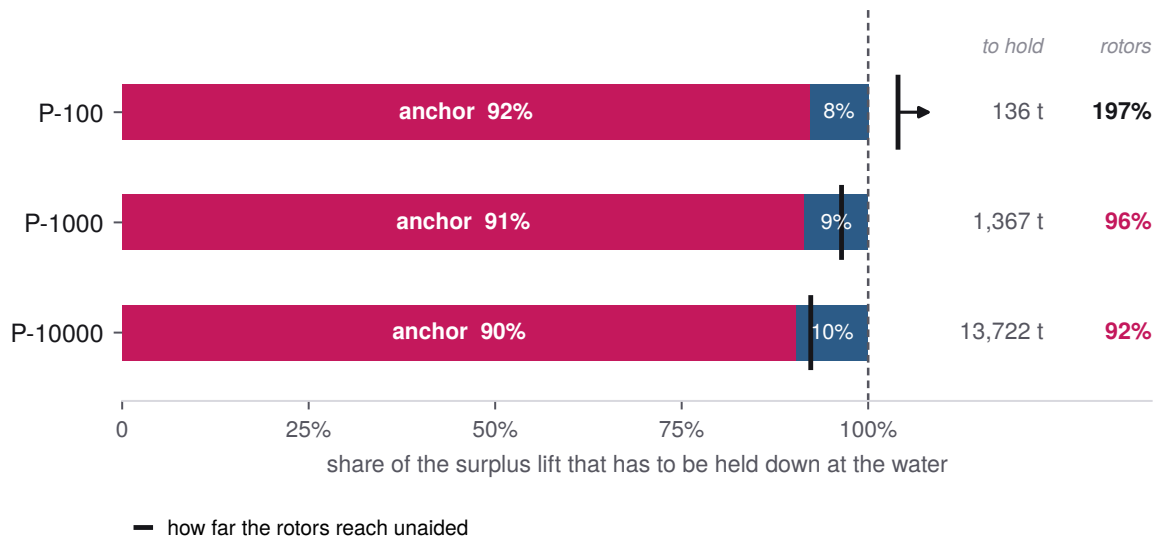


Figure 3. What has to be held down at the water, and what holds it. The bar is the job; the tick is how far the rotors reach unaided.

6.2 Three ways out, costed

1. **Retain water as ballast.** Directly reduces delivery, which is the metric.
2. **Make ballast from air.** Liquefy nitrogen on the return leg. The plant is real and it is sized for something else (§7.2): at cycle rate it makes 21.12 t against a need of thousands. Making the whole 12,400 t as LN2 costs about **475 MWh**, eleven times the entire cycle.
3. **Never enter the dense air.** Hover at 1,350 m and fill through a long hose. Costed at **44 MWh** a cycle in pump work, with a 2 m bore at 140 bar at the pod.

6.3 What was actually chosen: borrow the lake

Lower a cable with a collapsible bag, fill it, winch it clear of the surface. Water hanging on a line is downward force at the cost of the lift needed to break the surface — 15 m of it.

	P-100	P-1000	P-10000
Bag	125 t	1,250 t	12,400 t
Cable	350 m	600 m	850 m
Pull	1.2 MN	12.3 MN	121.6 MN
Energy cost	0.006 MWh	0.060 MWh	0.596 MWh

The leverage is in the exponent. Induced rotor power goes as $\text{thrust}^{1.5}$, so moving load off the

rotors pays superlinearly: carrying 90% of the hold on the bag drops the required rotor power from 1,748 MW to 52.3 MW, a 97% reduction, and the letdown term from 34.20 MWh to 1.420 MWh.

That is why the bag is sized to do the *whole* descent rather than to cover the 1,056 t shortfall that revealed the problem. The shortfall was the symptom; the exponent was the finding.

Three consequences worth stating plainly:

- **Every class carries one, including the P-100 whose descent closes without it.** Removing its bag costs 1.526 MWh a cycle against 1.253: a 34% saving on a class that does not need the mechanism. A bucket is cheaper than thrust everywhere.
- **The mechanism cannot be over-sized.** The most water a ship can lift out of a lake is its own surplus lift; a bag equal to the surplus leaves the hull neutral. The physics supplies the ceiling.
- **A cable is a much better thing to hang than a hose.** 121.6 MN is about 440 mm of UHMWPE massing 125 t — 1.25% of payload in rope, and **not charged as dry mass anywhere in this model.**

6.4 What the anchor forces on the flight profile

The mechanism only works if the ship is not moving. A 12,400 t bag on an 850 m cable at 130 km/h is not an anchor, it is an accident. The model therefore requires: a dead stop before the cable goes out; no yaw while the lines are down; and departure permitted as soon as the pumps clear the water. This costs the 5-minute approach phase and buys the mechanism.

7. The energy ledger

7.1 The budget

P-10000, 15 km one way, balanced, still air. Printed from the model, not transcribed:

Term	MWh	share
RETURN_TRANSIT (drag $\times$ 0.55 + cryogenic plant)	14.705	34.2%
WATER_FILL (pumping 10,000 t up 300 m)	10.900	25.3%
other (hotel + manoeuvring drag)	10.689	24.8%
OUTBOUND_TRANSIT (drag, loaded)	9.459	22.0%
letdown (rotor work, with the anchor deployed)	1.420	3.3%
anchor (lifting the bag 15 m)	0.596	1.4%
recovery (nitrogen store, credited back)	-1.900	-11.0%
total	**45.869**	

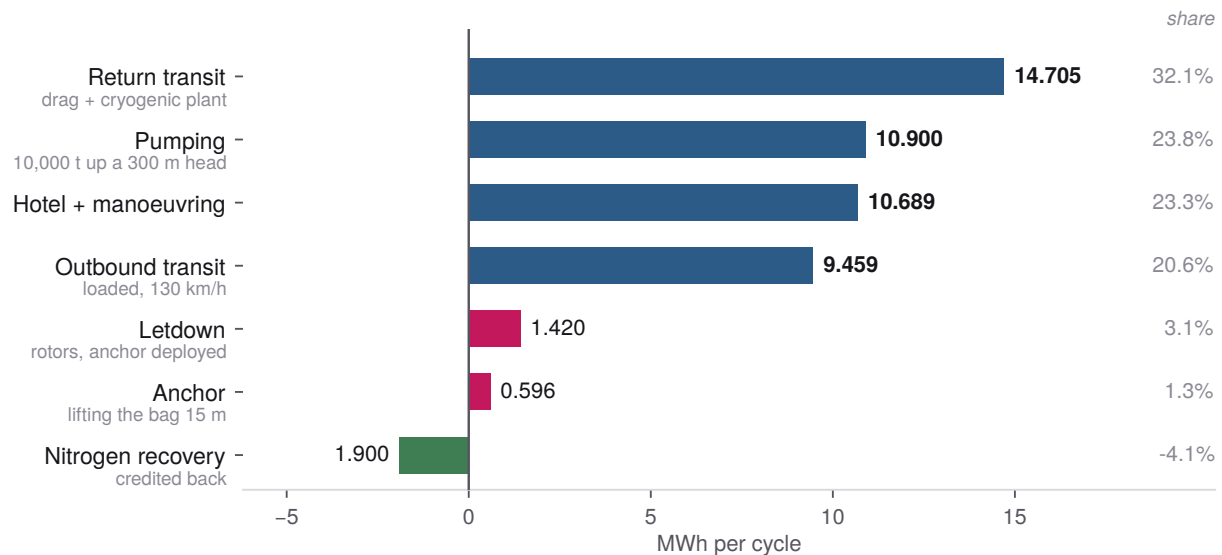


Figure 4. The cycle ledger, printed from the model rather than transcribed. The anchor is the smallest positive line, and it is the reason the letdown is the second smallest.

4.59 kWh/t delivered, against 7.40 for the P-1000 and 12.53 for the P-100. Larger is cheaper per tonne, as the square-cube law demands.

The bottom two rows were one row until 2026-08-09, and merging them was hiding an error: the recovery was netted against the pump bill under a $\max(0, \dots)$, so on the smaller classes the recovery exceeded the pumping and the excess was *deleted*, taking the whole pump bill with it. Splitting them moved the P-100 from 1.308 to 1.005 MWh. **The lesson generalises: the two most consequential errors found in this model were both accounting, not physics.**

7.2 What the cryogenic plant is actually for

The nitrogen plant makes 21.12 t per cycle against a 15,500 t tank. As cycle ballast it is three orders of magnitude short, and saying so is more useful than defending it.

Its real job is unpowered recovery. The tank is sized so that an empty hull can be made heavy enough to **land with no rotor authority at all**, at ground level rather than at the ceiling — 14,456.1 t of ballast to sink an empty P-10000 at 1,000 m MSL. Filling it costs about 6,505 MWh: 2.7 days at rated plant power, 12.9 days on solar. That is a rescue that takes days, and it should stop being confused with a cycle that takes an hour.

7.3 Generation, and the deficit

	solar	per cycle	spend	deficit	endurance
P-100	0.27 MW	0.15 MWh	1.253	1.10	10.4 h
P-1000	1.26 MW	0.74 MWh	7.399	6.66	10.6 h
P-10000	5.40 MW	4.10 MWh	45.869	41.77	36.3 h

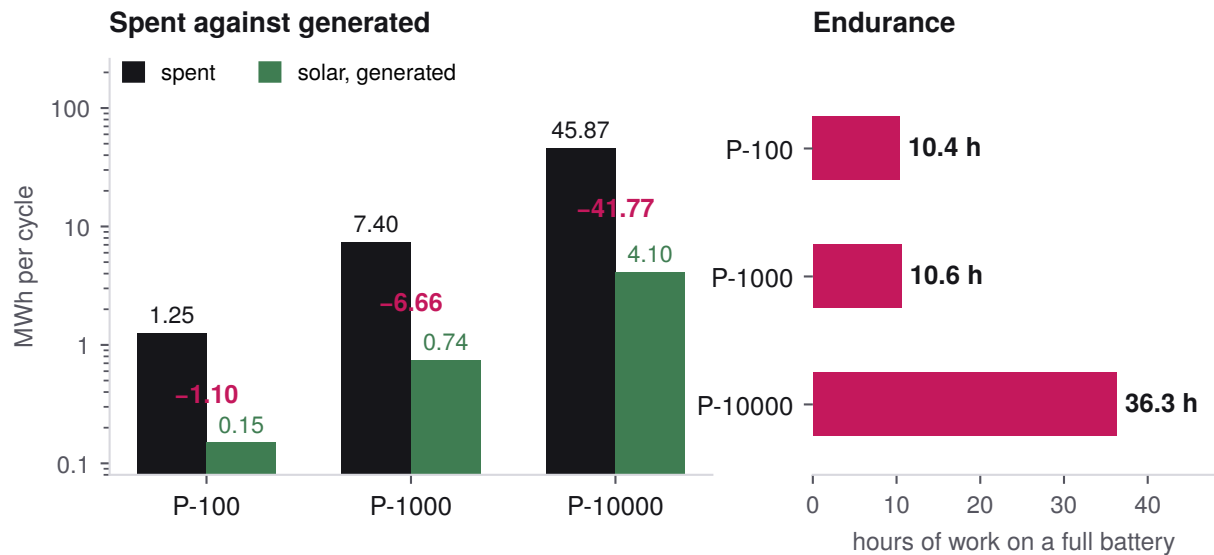


Figure 5. What each hull spends against what its skin makes, and how long a full battery lasts.

Every hull runs a deficit every cycle, and this is the project’s central public conclusion: without an energy import chain the fleet is a battery being spent. It is not a perpetual machine and does not claim to be.

Two defects attack this table from opposite directions. The generators each class advertises — 8, 40 and 150 MW — supply thrust authority and no energy at all, which understates generation. And the solar figure is credited at 200 W/m^2 , which overstates it by about $3.8 \times$ (§8.2). Correcting both moves the conclusion in the same direction it already points.

8. What is wrong with this model

Thirteen entries are documented in `docs/OPEN-QUESTIONS.md`. Six were found by inspection, two by auditing all 89 published claims against the code, and four by reading the sources. The four from the sources are the serious ones and they are given here in full.

8.1 The dry-mass budget fails twice, independently — the largest open question

$\text{dryT} = \text{payloadT}$ implies a hull-average density of 0.455 kg/m^3 : 10,000 t of everything-that-is-not-water inside $22,000,000 \text{ m}^3$.

- **Jenett et al. 2019** (NASA NTRS) — the structural precedent this project cites — gives a bare discrete-lattice shell at 0.508 kg/m^3 in its own Table 2, at every radius. That is 12% over our *entire* dry allowance, before skin, joints, rotors, tanks, batteries, or the 125 t of anchor cable §6.3 admits is uncharged.
- **Metlen & Palazotto 2013**’s only real-materials vacuum-lift design has a structure-to-buoyancy ratio of 0.94 — structure alone consuming what we allocate to structure and payload together.
- **Akhmeteli & Gavrilin 2021** publish a payload fraction of 0.1 against our implied 0.5 , on a shell of 1.16 kg/m^3 .
- Independently, every class carries **0.2 MWh of battery per tonne of dry mass**, requiring 200 Wh/kg at *pack* level with nothing left for anything else. NASA flew 149 Wh/kg on X-57; Lvovich (2020) sees no clear path past 500 Wh/kg at pack level, where the battery still consumes 40% of the budget.

Four sources, four different objections, one conclusion: **the mass premise is not supported by the literature this project itself cites.** Everything in §§5–7 is conditional on it.

8.2 The solar skin needed 76% conversion efficiency — FIXED 2026-08-09

`state.js` credited 200 W/m^2 of *electrical output*, continuously, in **five separate files**. NRCan's dataset gives $6.34 \text{ kWh/m}^2/\text{day}$ mean July horizontal insolation across eight BC interior fire-belt towns — 264 W/m^2 incident, day-averaged. 200 out of 264 is **76% conversion**, three and a half times the best cell ever made in a laboratory.

It is `CFG.solarWPerM2 = 45` now: 264×0.21 flexible module $\times 0.81$ for curvature, cell temperature, soiling and conversion, applied to a **projected** area — `solarM2` is 80–87% of each hull's plan ellipse, so the curvature is paid for once, in the area.

	solar	per cycle	deficit	endurance
P-100	1.20 → **0.27 MW**	0.68 → **0.15 MWh**	0.32 → **1.10**	35.4 → **10.4 h**
P-1000	5.60 → **1.26 MW**	3.30 → **0.74 MWh**	3.09 → **6.66**	22.9 → **10.6 h**
P-10000	24.00 → **5.40 MW**	18.20 → **4.10 MWh**	24.81 → **41.77**	61.1 → **36.3 h**

Three things this exposed are worth more than the correction. The constant was duplicated five times and wrong in every copy — there is one now, plus one on the far side of the `3d/` boundary and a parity test that reads both. **No published figure moved**, because generation is not in `planCycle`'s ledger: cycle energy, throughput and kWh/t came out bit-identical across a $4.4\times$ change to the vehicle's power supply, which is defect 6 in §8.5 stated as sharply as it can be. And the honest number strengthens the project's conclusion rather than weakening it.

Still open: 45 W/m^2 is a 24-hour average — right for energy over a cycle, wrong for power at an instant. There is no sun at 03:00 and the storage gauge draws it anyway.

8.3 `rtLN2` returned more work than the nitrogen contains — FIXED 2026-08-09

`rtLN2 = 0.50` against `eLN2 = 0.45` kWh/kg recovered **225 kWh per tonne** of liquid nitrogen. The physical exergy of LN2 at 1 bar against a 288 K ambient is **173.4 kWh/t** (Arnaiz-del-Pozo et al. 2020; corroborated at 205–214 kWh/t under more favourable assumptions). The term returned $1.3\times$ the work available in the liquid, before any turbine or generator efficiency. Not an optimistic efficiency — a violation, and it was on the page.

It is ****0.20**** now: 90 kWh/t, 52% of the exergy, about what a cryogenic expander returns with no external heat source. `E.recovery` fell from -4.751 to -1.900 MWh and the P-10000's cycle rose to 45.869. Two tests enforce $\text{rtLN2} \times \text{eLN2} \times 1000 \leq 173.4$, one on each copy of the constant, because a second law is not a tuning bound.

8.4 The drop may not arrive, and tonnes may be the wrong metric

The US Forest Service's 2022 assessment states that a drop released 1,000 ft above ground/vegetation "would completely dissipate". `ALT.drop` is 450 m — **1,476 ft** — raised for hull-clearance reasons that are sound, with the delivery consequence never costed. The model has no droplet physics: it moves tonnes from a tank to a coordinate.

And AFUE — 27,611 observed drops — reports probability of success **0.56 without ground engagement against 0.72 with it**, the modal outcome without ground crews being *not effective*. AFUE never counts tonnes. This project’s headline metric is tonnes per hour.

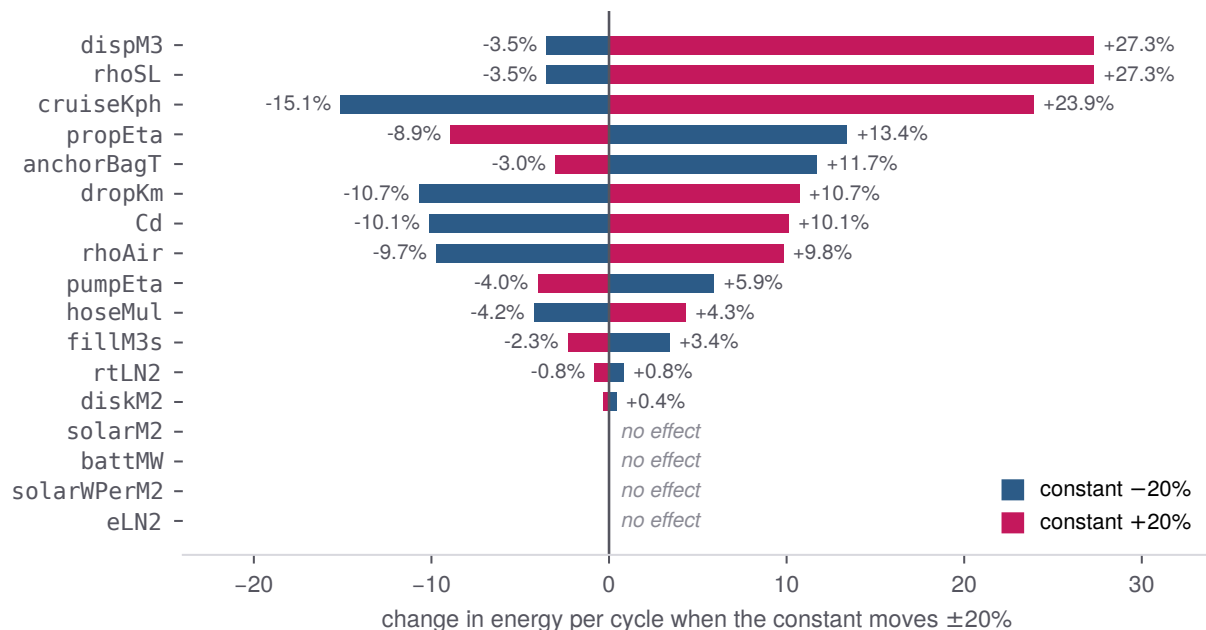


Figure 6. Every constant moved $\pm 20\%$, measured rather than asserted. The top of the table is aerodynamics and speed; the two constants corrected on 2026-08-09 now move the headline by 0.8% and 0.0%.

8.5 The other nine, in one line each

#	Defect	Cost
1	Lift bought at sea level, spent at altitude	fixed 2026-08-09; hulls grew 22.2%
2	Two power models disagree	planCycle vs integrated stateAt: 2.83x on
3	An unexplained 0.2 sets the letdown window	1.42 MWh today; was 34.20 before the anchor
4	Retained ballast is dead code; the plant is inert in-cycle	ballast fixed ; plant restated (\$7.2)
5	Esri basemap tiles used outside their terms	not physics; blocking for publication
6	Generators supply peak power but no energy	150 MW uncounted on the P-10000 — and \$8.2
7	32 of 89 published figures rest on unjustified constants	two inline drag multipliers are worth 16.9 MW
8	diskM2/battMW answer a superseded constraint	battMW $\pm 20\%$ now moves every figure by 0.0
12	Rotor wash over a water surface is unmodelled	Suter (2005); a 121.6 MN anchor hangs in it

Defect 7 deserves a sentence of its own. The audit found that the constants nobody had questioned move the headline more than the defects everyone had: the two undocumented drag multipliers on the return and manoeuvring phases are worth 16.9 MWh a cycle, while defect 3 — flagged in the README since the first commit — is worth 1.4.

9. How to check any of this

```
" git clone ... && cd airships && make check "
```

- `make golden` re-runs 151 class/mode/distance/wind combinations and diffs every output against `tests/golden/seed7-snapshot.json`.
- `make test` runs 206 tests in 39 suites. Two are marked `knownFail` and **fail on purpose** — a defect is not allowed to lose its excuse quietly.
- `make test-node` runs a further 103. Until 2026-08-09 it printed "SKIPPED — no node here" on the machine this is developed on, so those 103 were only ever executed by CI and two breaks were found by a red badge after a push. They run in a browser now, against the same files, and `make check` covers them.
- `make factsheet` regenerates `research/figures.json` from the live model in a real browser; `tools/check_figures.py` then fails the build if any number in this paper disagrees with it. Every figure marked above passed that check at the commit that carries this file.
- `?seed=7&data=snapshot` makes any run on the live site exactly reproducible.

An independent Python replication reproduces the golden snapshot exactly.

10. What would have to be true

In descending order of how likely it is to kill the concept:

1. **A discrete-lattice vacuum shell at $\leq 0.455 \text{ kg/m}^3$ including everything.** Nothing in the literature is there. This is the concept's single point of failure and no amount of care in the rest of the model substitutes for it.
2. **Batteries at 200 Wh/kg pack-level with the rest of the vehicle free.** Not available.
3. **A 12,400 t bucket on a 440 mm cable, and a hull that can hold station over water while it hangs there.** The mechanism is a scaled Bambi bucket at 1,265 times the largest ever built (9,800 L) — the principle is 43 years old and the engineering is not. Pendulum dynamics under an 876 m hull are not modelled.
4. **A drop from 450 m that arrives as water rather than as mist.** Currently contradicted by the USFS's own guidance.
5. **An energy import chain.** The fleet is a battery being spent, and the 2026-08-09 solar correction made that sharper rather than softer: the smallest class now has 10.4 hours of work in it and the largest 36.3. Nothing here changes that; §7.3 is where it is costed.

If (1) fails, the rest is a well-tested model of a vehicle that cannot exist. That is why it is first on the list and why §8.1 is written the way it is.

What the two corrections of 2026-08-09 say about the other eleven. Both were found by reading a source rather than by running the model, both had been on the page for the life of the project, and neither was catchable by any test that existed — one because generation is absent from the ledger, the other because a test was asserting the violation as correct behaviour. The eleven still open should be read in that light: the list is what has been checked, not what is wrong.

11. Availability

Model, tests, sources and defect list: github.com/pinkrobotics/airships. Live simulation: pinkrobotics.ca/airships. The literature and the nine sources against us, written for a reader who has not cloned anything: pinkrobotics.ca/research. Source catalogue: `research/sources.json` — 73 entries, 29 redistributable PDFs with provenance, 12 written

notes, 9 sources that contradict us. Claim audit: `research/evidence-map.md`.

Every figure marked in this document is generated by `tools/figures_dump.js` from the model at defaults and verified by `tools/check_figures.py`. Figures from cited sources are written plainly and accounted for in `research/sources.json`.